

Bruna Zamith Santos, PhD Candidate

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MAIN AREAS OF INTEREST

- **Data Science, Machine Learning, Research & Development, Software Engineering**

PROFESSIONAL EXPERIENCE

- **iFood** Remote - Brazil
Staff Data Scientist Dec. 2025 – Current
 - Develop recommendation models and bidding optimization systems for Retail Media and Ads products.
 - Own end-to-end ML development: business framing, experimentation, model training, deployment, monitoring, and optimization.
 - **Technologies:** AWS, Python, SQL
 - **Domains:** Recommender Systems, Optimization, GenAI, A/B Testing, MLOps
- **Amazon** Remote - Brazil
Applied Scientist II Feb. 2024 – Dec. 2025
Data Scientist II Mar. 2022 – Jan. 2024
 - Own end-to-end supervised ML for Catalog and Supply Chain: business framing, data/feature pipelines, model training, offline and online evaluation, deployment, monitoring, and MLOps.
 - Delivered models credited with \$50M+ incremental revenue and enabling 1.3M+ new catalog product launches through scalable decisioning services and data products.
 - Partnered with PM/Eng/Ops to prioritize a multi-quarter roadmap, negotiate trade-offs (latency/cost/quality), and unblock adoption across LatAm businesses.
 - Program Owner, GenAI capacitation: designed curriculum and delivered trainings and office hours for different job roles across 12 marketplaces (10k+ employees); measured adoption and learning outcomes.
 - Published 2 internal Amazon papers (applied ML/GenAI); evangelized best practices in experimentation, evaluation, and productionization.
 - **Technologies:** AWS, Python, SQL, Typescript
 - **Domains:** GenAI, Forecasting, Classification, A/B Testing, MLOps
- **Amazon** Sao Paulo, SP - Brazil
Software Development Engineer II Dec. 2021 – Feb. 2022
Software Development Engineer I Jul. 2020 – Nov. 2021
Software Development Intern Jan. 2020 – Jul. 2020
 - Built and operated large-scale front-/back-end services integrating with Amazon core systems to improve seller & buyer experiences.
 - Led design reviews, API contracts, performance testing, reliability, and on-call for high-throughput workloads.
 - Shipped resilient, well-instrumented services adopted by core retail flows in Brazil.
 - **Technologies:** AWS, Java, Python, Typescript
 - **Domains:** Distributed Systems, APIs, Observability, Reliability
- **Serasa Experian** Sao Carlos, SP - Brazil
MIS Intern Aug. 2019 – Dec. 2019
 - Delivered Finance dashboards and automated data pipelines; improved reporting accuracy and turnaround time by 40%.
 - **Technologies:** SQL, SAS, Tableau
 - **Domains:** Business Intelligence, Data Engineering

EDUCATION & RESEARCH

- **Federal University of Pernambuco (UFPE)** Remote — Brazil
PhD in Artificial Intelligence Feb. 2025 – Present
 - **Topic:** Explainable AI (XAI) for Time Series Classification.
 - **Technologies:** Python (PyTorch/NumPy/Pandas), LaTeX
 - **Domains:** XAI, Time Series, Instance Hardness
 - **GPA:** 10/10
- **Federal University of Sao Carlos (UFSCar)** Remote - Brazil
Masters in Artificial Intelligence Aug. 2021 – Dec. 2023

- **Project:** Climate Variables Forecasting & Forest Fire Risk Classification in the Brazilian Pantanal. Pantanal faces recurrent fires driven by climate; existing indexes are biome-agnostic, limited in variables, and lack multi-day predictions.
- **Approach:** Built a two-stage ML system: (1) forecast climate variables; (2) classify daily fire-risk levels using predicted covariates. Compared exhaustive hyperparameter search to Genetic Algorithms (GA) for multi-objective optimization.
- **Results:** System outperformed standard fire-risk indexes on key classes and enabled forward-looking risk.
- **Technologies:** Python, R
- **Domains:** Time Series, Forecasting, Classification, Genetic Algorithms
- **GPA:** 10/10

• Katholieke Universiteit Leuven (KU Leuven)

Kortrijk, Flanders - Belgium

Research Intern

Sep. 2017 – Dec. 2017

- **Project:** Predicting Protein Functions via Interaction Prediction. Funded by Sao Paulo Research Foundation (FAPESP).
- **Approach:** Modeled protein function prediction as Hierarchical Multi-label Classification (HMC) over interaction networks; implemented data processing and evaluation pipelines.
- **Outcome:** Delivered reproducible code and benchmarks; informed subsequent HMC research at BioMaL/KU Leuven.
- **Technologies:** Java
- **Domains:** HMC, Bioinformatics

• Federal University of Sao Carlos (UFSCar)

Sao Carlos, SP - Brazil

Bachelor in Computer Engineering

Mar. 2015 – Jul. 2020

- **GPA:** 8.75/10

PROGRAMMING SKILLS

- **Languages:** Python (6 years); Java (5 years); SQL (3 years); Typescript (2 years); R (2 years); C++ (6 months)
- **Others:** AWS, Git, Linux, Latex

LANGUAGE

- **Portuguese:** Native
- **English:** Fluent (Cambridge English Certificate ESOL Intl)

PRESENTATIONS, PROCEEDINGS, AND PAPERS

- **TSCFEval: A Model-Agnostic Framework for Evaluating Time Series Classification Counterfactuals** B. Zamith Santos, M. F. Andrade Lira, R. Cerri. 2026. In *Explainable Artificial Intelligence (XAI World Conference)*.
- **A New Time Series Framework for Forest Fire Risk Forecasting and Classification** B. Zamith Santos, B. Soriano, M. Narciso, D. Furtado, R. Cerri. 2023. In *International Joint Conference on Neural Networks (IJCNN)*.
- **Predictive Bi-Clustering Trees for Hierarchical Multi-Label Classification** B. Zamith Santos, K. F. Nakano, R. Cerri, C. Vens. 2020. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)*.
- **Strategies for Selection of Positive and Negative Instances in the Hierarchical Classification of Transposable Elements** B. Zamith Santos, T. G. Pereira, K. F. Nakano, R. Cerri. 2018. In *Brazilian Conference on Intelligent Systems (BRACIS)*.
- **A Genetic Algorithm for Transposable Elements Hierarchical Classification Rule Induction** T. G. Pereira, B. Zamith Santos, R. Cerri. 2018. In *IEEE Congress on Evolutionary Computation (IEEE CEC)*.
- **A New Machine Learning Dataset for Hierarchical Classification of Transposable Elements** B. Zamith Santos, R. Cerri. 2016. In *National Meeting on Artificial and Computational Intelligence (ENIAC)*.
- **Decision Trees for Hierarchical Classification of Transposable Elements** B. Zamith Santos, R. G. Mantovani, L. Schietgat, C. Vens, R. Cerri. 2016. In *Proceedings of the 25th Belgian-Dutch Machine Learning Conference (Benelearn)*.

CERTIFICATIONS

- **Statistics and Optimization for Data Science and Operations Research (2026):** 60 hours online course provided by University of Sao Paulo (USP).
- **Recommender Systems Specialization (2025):** 80 hours online course provided by University of Minnesota.
- **Fundamentals of Probability and Statistics for Data Science (2025):** 30 hours online course provided by University of Sao Paulo (USP).
- **Practical Multi AI Agents and Advanced Use Cases with crewAI (2025):** 3 hours online course provided by DeepLearning.ai and crewAI.
- **Practical Time Series Analysis (2024):** 25 hours online course provided by The State University of New York (SUNY).
- **Generative AI with Large Language Models (2024):** 16 hours online course provided by DeepLearning.ai and Amazon Web Services (AWS).
- **Sequence Models (2022):** 38 hours online course provided by DeepLearning.ai.
- **Sequences, Time Series and Prediction (2022):** 23 hours online course provided by DeepLearning.ai.
- **Introduction to AWS (2020):** 5 hours online course provided by A Cloud Guru.
- **Version Control With Git (2019):** 12 hours online course provided by Atlassian.
- **Software Development Processes and Methodologies (2019):** 18 hours online course provided by University of Minnesota.